

$$62. \text{ Given exp. } = \left(\frac{5}{7} \times \frac{19}{13}\right) \div \left(\frac{19}{7} \times \frac{4}{13}\right) = \frac{5 \times 19}{7 \times 13} \times \frac{7 \times 13}{19 \times 4} = \frac{5}{4}.$$

$$63. \text{ Given exp. } = \frac{11}{4} \div \frac{8}{3} \div \frac{13}{12} = \frac{11}{4} \times \frac{3}{8} \times \frac{12}{13} = \frac{99}{104}.$$

$$64. \text{ Given exp. } = \frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{2}{3} \text{ of } \frac{5}{4}} = \frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{5}{6}} = \frac{2}{3} \times 3 = 2.$$

$$65. \text{ Let } \frac{4335}{x} \div \frac{15}{8} = \frac{289}{528}.$$

$$\begin{aligned} \text{Then, } \frac{4335}{x} &= \frac{289}{528} \times \frac{15}{8} \Leftrightarrow \frac{4335}{x} = \frac{289 \times 5}{176 \times 8} \\ \Leftrightarrow x &= \left(\frac{4335 \times 176 \times 8}{289 \times 5}\right) = 4224. \end{aligned}$$

$\therefore$ Missing digit = 2.

$$66. \text{ Let } \frac{16}{3} - \frac{11}{3} \div \frac{4}{3} \div x + \frac{16}{5} \div \frac{6}{5} = 7. \text{ Then,}$$

$$\frac{16}{3} - \frac{11}{3} \times \frac{3}{4} \times \frac{1}{x} + \frac{16}{5} \times \frac{5}{6} = 7$$

$$\Leftrightarrow \frac{16}{3} - \frac{11}{4x} + \frac{8}{3} = 7$$

$$\Leftrightarrow \frac{24}{3} - \frac{11}{4x} = 7$$

$$\Leftrightarrow \frac{11}{4x} = 8 - 7 = 1$$

$$\Leftrightarrow 4x = 11 \Leftrightarrow x = \frac{11}{4} = 2\frac{3}{4}.$$

$$67. \text{ Given exp. } = 9 - \frac{11}{9} \text{ of } \frac{36}{11} \div \frac{36}{7} \text{ of } \frac{7}{9} = 9 - 4 \div 4 = 9 - 1 = 8.$$

$$68. \text{ Let } \frac{5}{6} \div \frac{6}{7} \times x - \frac{8}{9} \div \frac{8}{5} + \frac{3}{4} \times \frac{10}{3} = \frac{25}{9}. \text{ Then,}$$

$$\frac{5}{6} \times \frac{7}{6} \times x - \frac{8}{9} \times \frac{5}{8} + \frac{3}{4} \times \frac{10}{3} = \frac{25}{9} \Leftrightarrow \frac{35}{36}x - \frac{5}{9} + \frac{5}{2} = \frac{25}{9}$$

$$\Leftrightarrow \frac{35}{36}x = \frac{25}{9} + \frac{5}{9} - \frac{5}{2} = \frac{10}{3} - \frac{5}{2}$$

$$\Leftrightarrow \frac{35}{36}x = \frac{5}{6} \Leftrightarrow x = \left(\frac{5}{6} \times \frac{36}{35}\right) = \frac{6}{7}.$$

$$69. \text{ Given exp. } = \frac{3}{4} \div \frac{9}{4} \text{ of } \frac{2}{3} - \left(\frac{3-2}{6}\right) \times \frac{10}{3} + \frac{5}{6}$$

$$= \frac{3}{4} \div \frac{9}{4} - \frac{1}{6} \times \frac{6}{5} \times \frac{10}{3} + \frac{5}{6}$$

$$= \frac{3}{4} \times \frac{2}{3} - \frac{2}{3} + \frac{5}{6} = \left(\frac{1}{2} - \frac{2}{3} + \frac{5}{6}\right)$$

$$= \left(\frac{3-4+5}{6}\right) = \frac{4}{6} = \frac{2}{3}.$$

$$70. \frac{\frac{7}{3} + 1\frac{1}{2} \text{ of } \frac{5}{3}}{2 + 1\frac{2}{3}} = \frac{\frac{7}{3} + \frac{3}{2} \text{ of } \frac{5}{3}}{2 + \frac{5}{3}} = \frac{\frac{7}{3} + \frac{5}{2}}{\frac{11}{3}} = \frac{29}{6} \times \frac{3}{11} = \frac{29}{22}.$$

$$\therefore \text{ Required answer } = \frac{29}{22} - \frac{1}{4} = \frac{58-11}{44} = \frac{47}{44} = 1\frac{3}{44}.$$

$$\begin{aligned} 71. \text{ Given exp. } &= \frac{\frac{1}{3} + \frac{3}{4} \left(\frac{6-5}{15}\right)}{\frac{5}{3} \text{ of } \frac{3}{4} - \frac{1}{5}} = \frac{\frac{1}{3} + \frac{3}{4} \times \frac{1}{15}}{\frac{5}{4} - \frac{1}{5}} \\ &= \frac{\frac{1}{3} + \frac{1}{20}}{\frac{25-4}{20}} = \frac{23}{60} \times \frac{20}{21} = \frac{23}{63}. \end{aligned}$$

$$\begin{aligned} 72. \text{ Given exp. } &= 1 - \left(\frac{\frac{5}{3} + \frac{7}{6}}{\frac{7}{2} - \frac{7}{2}}\right) = 1 - \left(\frac{5}{3} \times \frac{2}{7} + \frac{7}{6} \times \frac{2}{7}\right) \\ &= 1 - \left(\frac{10}{21} + \frac{1}{3}\right) = 1 - \frac{17}{21} = \frac{4}{21}. \end{aligned}$$

$$\begin{aligned} 73. \text{ Given exp. } &= \frac{5}{2} \text{ of } \frac{3}{4} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \frac{3}{2} \left[\frac{2}{3} - \frac{1}{3}\right] \\ &= \frac{5}{2} \text{ of } \frac{3}{4} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \left(\frac{3}{2} \times \frac{1}{3}\right) \\ &= \frac{15}{8} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \frac{1}{2} \\ &= \frac{15}{8} \times \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times 2 = \frac{5}{8} + 1 = 1\frac{5}{8}. \end{aligned}$$

$$\begin{aligned} 74. \text{ Given exp. } &= \frac{\frac{8}{7} - \frac{2}{3} + \frac{2}{5} \times \frac{25}{24}}{1 - \frac{1}{7} \left[\frac{1}{3} + \frac{2}{5} \times \frac{5}{3}\right]} = \frac{\frac{8}{7} - \frac{2}{3} + \frac{5}{12}}{1 - \frac{1}{7} \left[\frac{1}{3} + \frac{2}{3}\right]} \\ &= \frac{96-56+35}{84} = \frac{75}{84} \times \frac{7}{6} = \frac{25}{24} = 1\frac{1}{24}. \end{aligned}$$

$$\begin{aligned} 75. \text{ Given exp. } &= \frac{6}{\left(\frac{10}{3}\right)} \div \frac{4 - \frac{2}{\left(\frac{7}{2}\right)}}{\left(\frac{7}{2}\right)} - \frac{2}{5} \text{ of } \left\{\frac{6}{9} + \frac{1}{3}\right\} \\ &= \frac{6 \times 3}{10} \div \frac{4 - \frac{2 \times 2}{7}}{\left(\frac{7}{2}\right)} - \frac{2}{5} \text{ of } 1 \\ &= \frac{9}{5} \div \frac{\left(4 - \frac{4}{7}\right)}{\left(\frac{7}{2}\right)} - \frac{2}{5} = \frac{9}{5} \div \left(\frac{24}{7} \times \frac{2}{7}\right) - \frac{2}{5} \\ &= \frac{9}{5} \times \frac{49}{24} - \frac{2}{5} = \frac{147}{80} - \frac{2}{5} = \frac{147-32}{80} = \frac{115}{80} = \frac{23}{16} = 1\frac{7}{16}. \end{aligned}$$

$$\begin{aligned} 76. \text{ Given exp. } &= \left(\frac{45}{8} \times \frac{7}{45}\right) \text{ of } \left(\frac{73}{11} \times \frac{8}{73}\right) + \frac{8}{9} \left(\frac{25}{11} + \frac{13}{22}\right) \text{ of } \frac{3}{5} \\ &= \frac{7}{8} \text{ of } \frac{8}{11} \div \left(\frac{8}{9} \times \frac{63}{22}\right) \text{ of } \frac{3}{5} \\ &= \frac{7}{11} \div \frac{28}{11} \text{ of } \frac{3}{5} = \frac{7}{11} \div \frac{84}{55} = \frac{7}{11} \times \frac{55}{84} = \frac{5}{12}. \end{aligned}$$